Software Requirements Specification

for

Medical Center

Version 1.0 approved

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10/1/2021

Revision History

Name	Date	Reason For Changes	Version
-	-	-	-

1. Introduction

1.1 Purpose

This is the first version of the medical center system. To fully manage small medical centers in terms : of facilitating and managing patient reservations, keeping their data from loss, facilitating the financial management of the medical center ,and facilitating dealing with the process of managing employees and all their affairs

1.2 Intended Audience and Reading Suggestions

This is SRS specially written for developers, software marketing personnel, and medical center owners who are looking for a system for their medical center that matches their needs and requirements. Therefore, we recommend that the reader first read the introduction in order to clarify the idea of the system in general, then look at the structure of the system in terms of the policy of managing orders, databases and restrictions for the users of the system

1.3 Product Scope

This system is limited to small medical centers. It has several goals and benefits, the most prominent of which is facilitating dealing with patients in terms of arranging their appointments, recording their data, adding their reports in the system and preserving them from damage, and is also useful in facilitating dealing with employees in terms of their salaries. Among the objectives of this system is to facilitate financial procedures and transactions in the medical center

1.4 References

1- Software Requirements Specification, Computer Information Systems Program, Janusz Zalewski, Ph.D. College of Business,Florida Gulf Coast University [visited in 9/1/2021].
2- Software Engineering: A Practitioner’s Approach, Seventh Edition, 2010[visited in 7/1/2021].
3- Software Engineering, Ninth Edition, Ian Sommerville, 2011[visited in 9/1/2021].
4- Software project management / Bob Hughes and Mike Cotterell, London[visited in 9/1/2021].

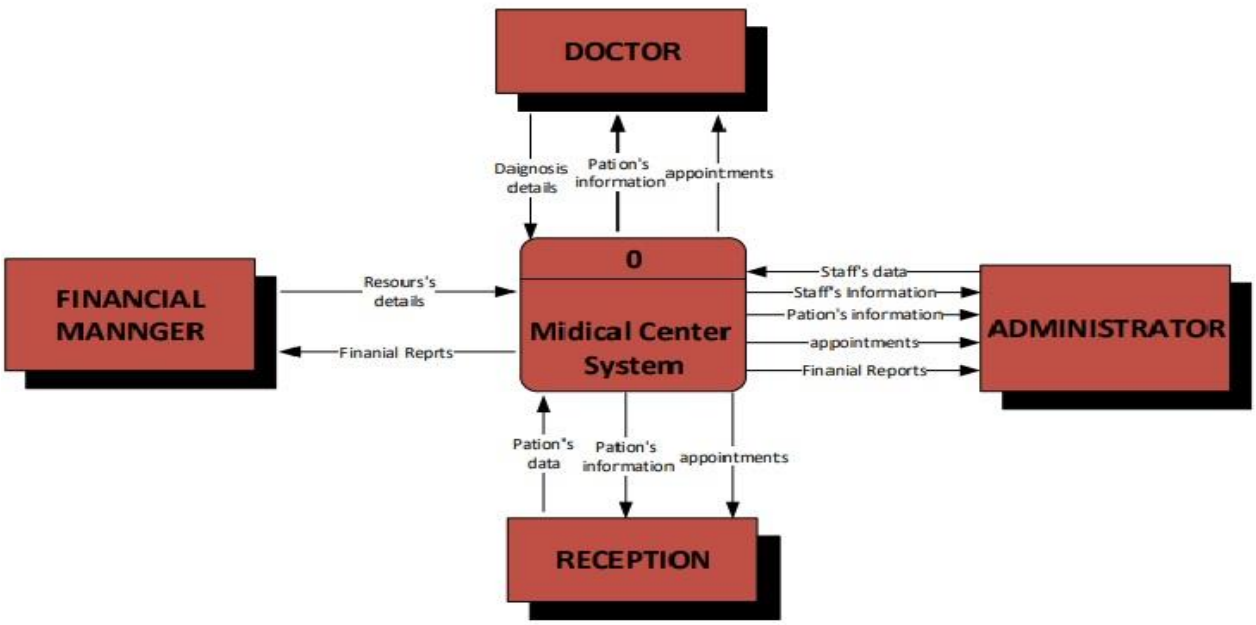
2. Overall Description

2.1 Product Perspective

This system will replace some of the current systems for managing medical centers, due to its ease of use, its high efficiency, the multiplicity of its tasks and the new tasks it contains
It is similar to some medical centers systems, but it has some new functions that facilitate the work

2.2 Product Functions

In this part, the functions will be explained in general to the users of the system with a context diagram display that facilitates the understanding of the users' functions of the system.
We will start with the system administrator, who can fully control the system, and these are some of its main functions: adding employees and entering their data to the system, displaying employee data, displaying appointments and displaying financial reports. As for the doctor, he can display appointments, patient information, and add a diagnosis to the patient. As for the receptionist, he can add and display appointments, patient data, and add data for a new patient. Finally, the financial manager has the authority to add resources to the system and their financial details, and he can also print financial reports.



figure(1), the main functions that the users of the system can do are explained

2.3 User Classes and Characteristics

Who are the system users ?? What are the features that they should possess ??
Four users can control this system: the admin, the doctor, the financial manager, and the receptionist .Each of them has its own characteristics. As the doctor has authority in the system that speed up his work, such as: reading patient information and adding the diagnosis. As for the admin, he has full authority to use the system and he must be used in the right way. The financial manager should be a person with experience in accounting and who has knowledge of using computers in his work. The receptionist must be a person quickly entering data in order not to overcrowd patients who want to register

2.4 Operating Environment

The environment that the system needs to function properly is not complicated and it is not very expensive, as its cost is moderate compared to most similar systems.
As this system can work on most Windows systems, but it is better to use Windows 7 or Windows 10, and also this system does not require very large RAM, as it works efficiently in any medium-efficient RAM and it is preferable to use this system in a Core i5 processor and more so that the system works with the required efficiency . In short, this system does not significantly affect the efficiency of the computer in terms of the processor and others, because the processes inside it are very simple and not the large complex that consumes the processor unit

2.5 Design and Implementation Constraints

The system works on more than one user, so we will face complications in the system, but not the great complications, for example, that each user has permeation and no user should interfere with the permeation of the other that are not shared by them, especially in the sensitive operations of the system such as controlling financial management, adding and deleting patients, etc. Operations are somewhat sensitive and there should be no chaos in controlling them, and also during the installation of the system, we must make sure that the computer is working well and also make sure that the devices are connected to the local network for interaction between users of the system in sensitive and emergency situations in a safe manner and also check the database program It works properly for all users so that there are no problems. There are complexities and limitations face us in designing and installing

2.6 User Documentation

An icon will be added for system users. It works like a user guide, as it helps the user in using the system, and can be referred to if the user encounters a problem

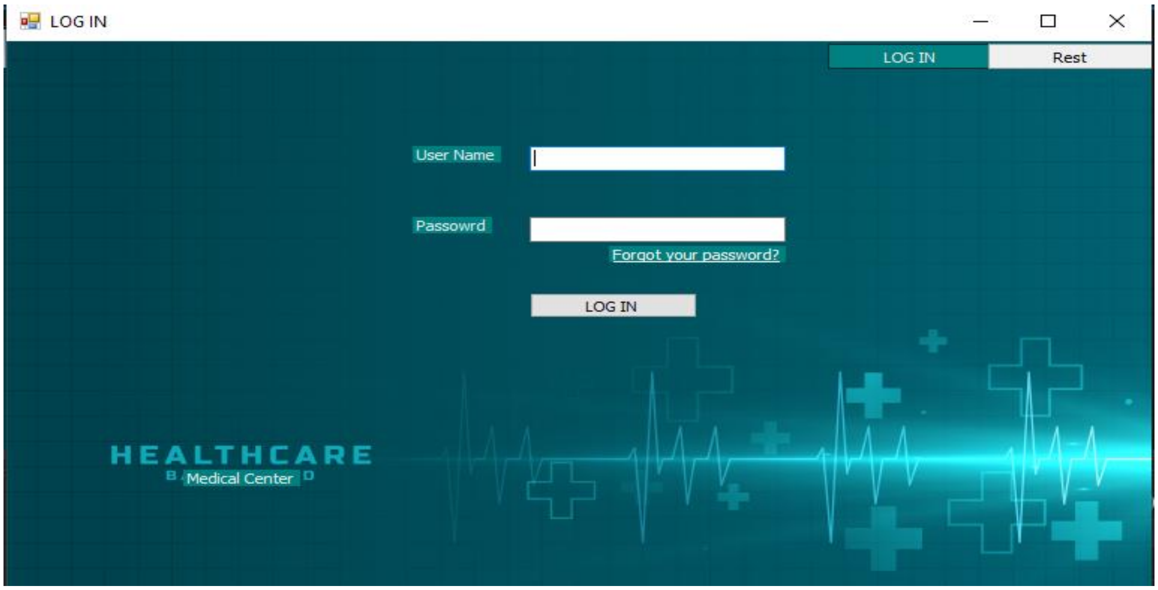
2.7 Assumptions and Dependencies

When the system is running, we will need a local network to communicate between users of the system or to share data between them. If there is a defect in this network, we will have problems in the system. It is possible in the future to develop this network by introducing the Internet to it so that patients can communicate with the medical center online or so that the admin can control the system remotely.

3. External Interface Requirements

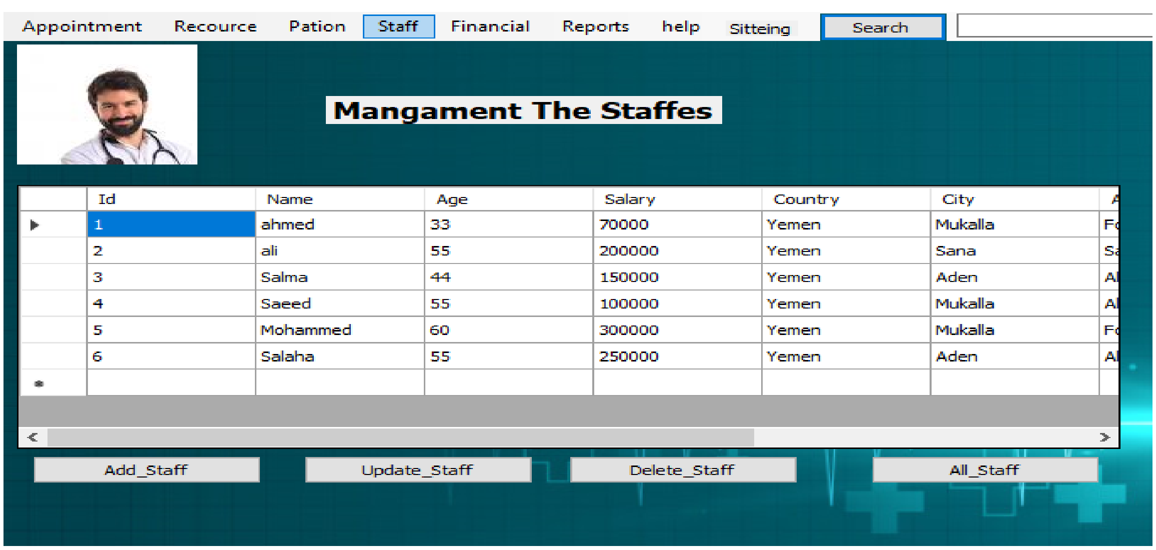
3.1 User Interfaces

In the following pictures, some of the interfaces used in the system are shown, which will interact with the users of the system

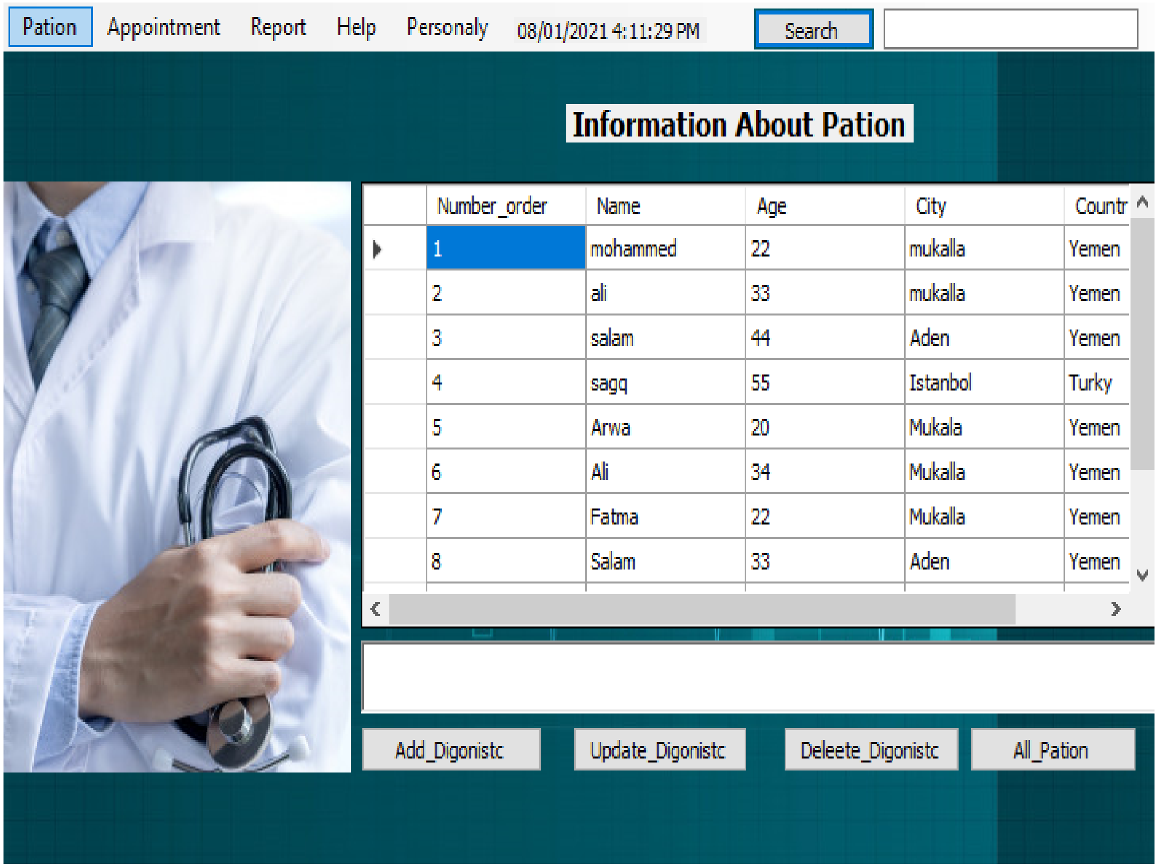


Figure(1) : log in interface

In Figure (1) this interface is specific to the login process for any user, as it contains the username and password
In Figure (2), this interface is especially for the admin, as it represents only one part of his work, which is the employee management, where he is exposed to employee data and he can add and remove an employee from the system and he can also modify their data



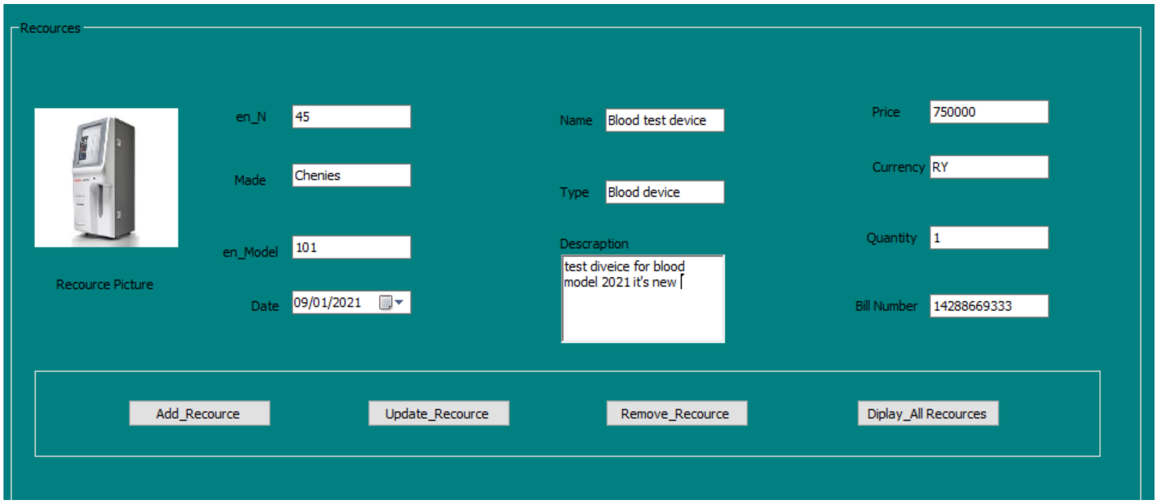
Figure(2) : add and delete and update and display staff from admin in Figure (3), this interface is for the doctor, where all patients are shown to him and through which he can search for a specific patient. It can also add, edit and delete any patient diagnosis. It can also display the data of all patients waiting to enter.



Figure(3) : add and delete and update and display diagnoses from doctor

In figure (4). this interface is especially for the receptionist, where he can add patients through it and also through it he can modify patient data and delete the patient, and as is clearly shown in front of him the arrangement of patients waiting to enter at the doctor determined by the receptionist at the top of the interface

Figure(4) : add and delete and update and display patios from reception in figure (5).this interface is for the financial manager, as he can add, delete and modify resources through this interface and also he can display all the resources in this interface



Figure(5) : add and delete and update and display recourse

4. System Features

4.1 Functional Requirements

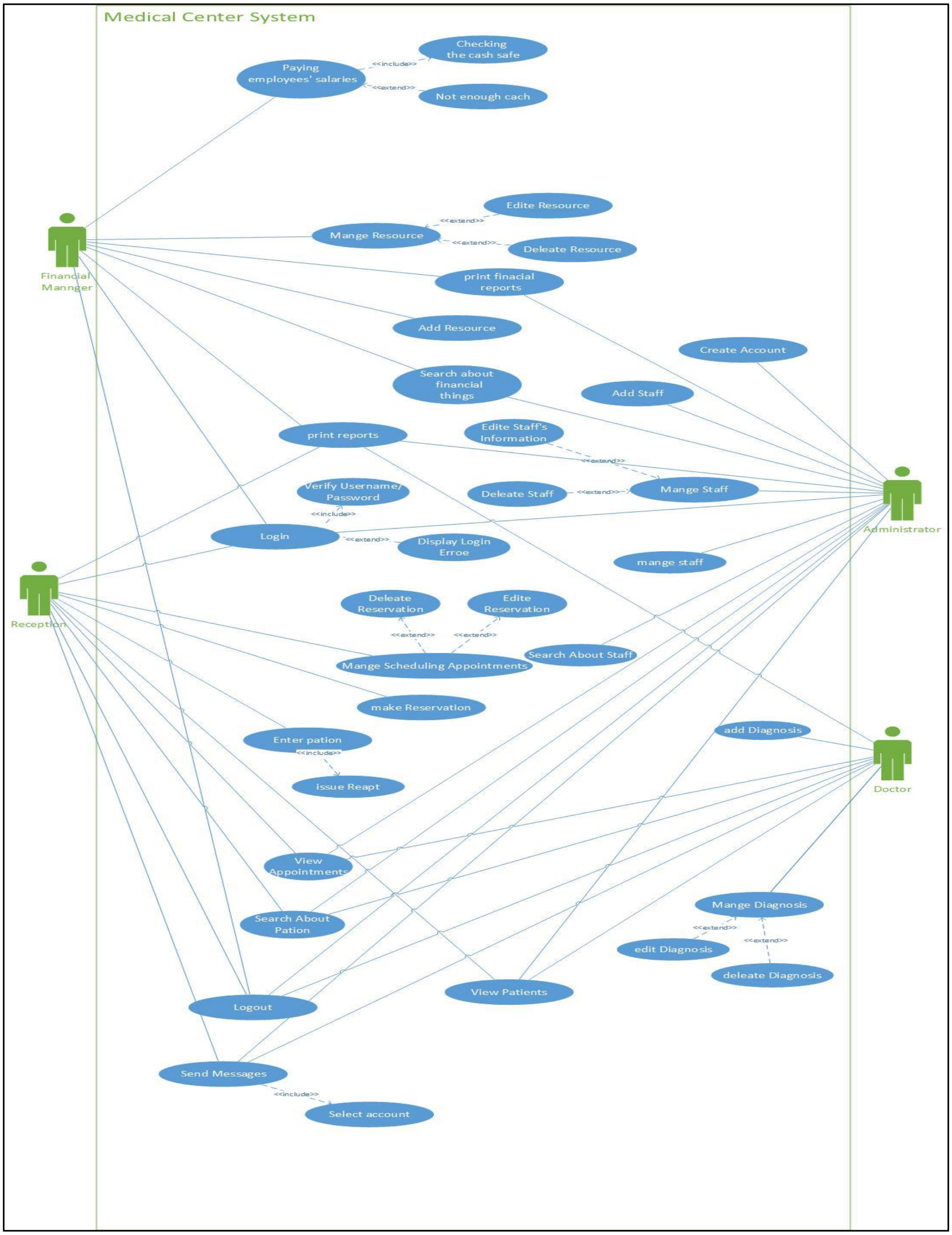
4.1.1 Use Case Glossary

Use case Name	Use -case description	Participating Actors
Create Account	This use case describes the process of creating new account in the system	Administrator
login	This use case describes the process of logging into the system	Administrator Doctor Reception Financial Manger
Add Staff	This use case describes the process of adding or register new staff in the system.	Administrator
Mange Staff	This use case describes the management process of staffs in the system, it can be by delete staff or update his information.	Administrator
Search About Staff	This use case describes the process of searching about staff information.	Administrator
Search about financial things	This use case describes the process of displaying the financial reports .	Administrator Financial Manger
Paying employees' salaries	This use case describes the event of paying the salaries of the staffs after checking the cash safe if there is enough cash.	Financial Manger
Print financial reports	This use case describes the process of printing the financial reports .	Financial Manger Administrator
Add Resource	This use case describes the process of adding the data and details of the new resource in the system.	Financial Manger

Use case Name	Use -case description	Participating Actors
Mange Resource	This use case describes the management process of the resources in the system, it can be by delete resource or update its data.	Financial Manger
Enter patient	This use case describes the process of entering a patient in the system, by adding their data in the system, with taking the money.	Reception
Mange patient	This use case describes the management process of the patients in the system, it can be by delete appointment or update its data.	Administrator
make Reservation	This use case describes the process of making a reservation for the patients, by adding their data in the system.	Reception
Mange Scheduling Appointments	This use case describes the management process of the appointments in the system, it can be by delete appointment or update its data.	Reception
View Appointments	This use case describes the process of displaying the appointments date and information.	Administrator Doctor Reception
Search About Patients	This use case describes the process of searching about patients information.	Administrator Doctor Reception
Send Messages	This use case describes the process of transferring the messages between the users in the system, by selecting the recipient .	Administrator Doctor Reception Financial Manger
View Patients	This use case describes the process of displaying patients information.	Administrator Doctor Reception
Add Diagnosis	This use case describes the process of adding diagnosis information.	Doctor

Use case Name	Use -case description	Participating Actors
Mange Diagnosis	This use case describes the management process of the diagnoses in the system, it can be by delete diagnosis or update its data.	Doctor
Logout	This use case describes the process of logging into the system	Administrator Doctor Reception Financial Manger

4.1.2 Use Case Diagram



Figure(6) : use case diagram

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Acceptability:
The system must be acceptable to the type of users for which it is designed . this means that it must be understandable ,useable and compatible with other system that they use.

Dependability and security:

system dependability includes a range of characteristic including reliability , security , and safety . dependable system should not cause physical or economic damage in the event of system failure . system has to be secure so that malicious users cannot access or damage the system.

Efficiency:

System should not make wasteful use of system resources such as memory and processor cycles Efficiency therefore includes responsiveness ,processing time ,resource utilization.

Maintainability:
System should be written in such a way that it can evolve to meet the changing needs of customers This is a critical attribute because system change is an inevitable requirement of a changing business environment.

5.2 Security Requirements

Users of the system must have their own accounts so that no other user can access it, so in this system there is a special database for user accounts that stores the user name and password for each account, and this is controlled by the administrator

5.3 Software Quality Attributes

This system is generally designed to adapt to a small medical center. It can also be used in more than one medical center, so that some operations are modified in order to adapt to that medical center. This system is designed in a way that the maintenance process is easy, and it is adapted to more than one possible medical center. Just as the development process for the system is not complicated because the processes and databases are designed in a way that the development is easy

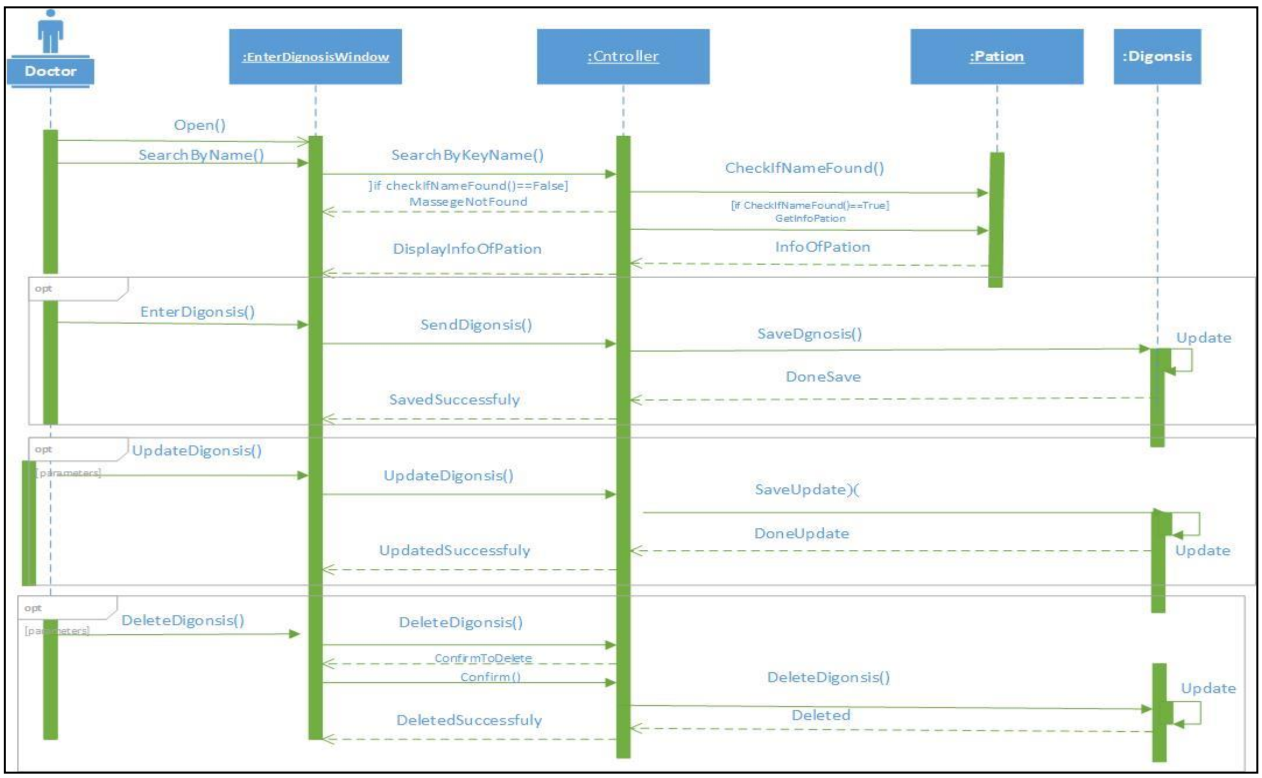
5.4 Business Rules

in this system, there are specific business rules, so that the administrator is the main controller of the system, and he is the one who controls the addition of users to the system, and he controls their removal and modification of their data. In this system, no user other than the admin can change the employees' data, and also the financial management, no user can modify anything related to finance except for the financial manager and admin

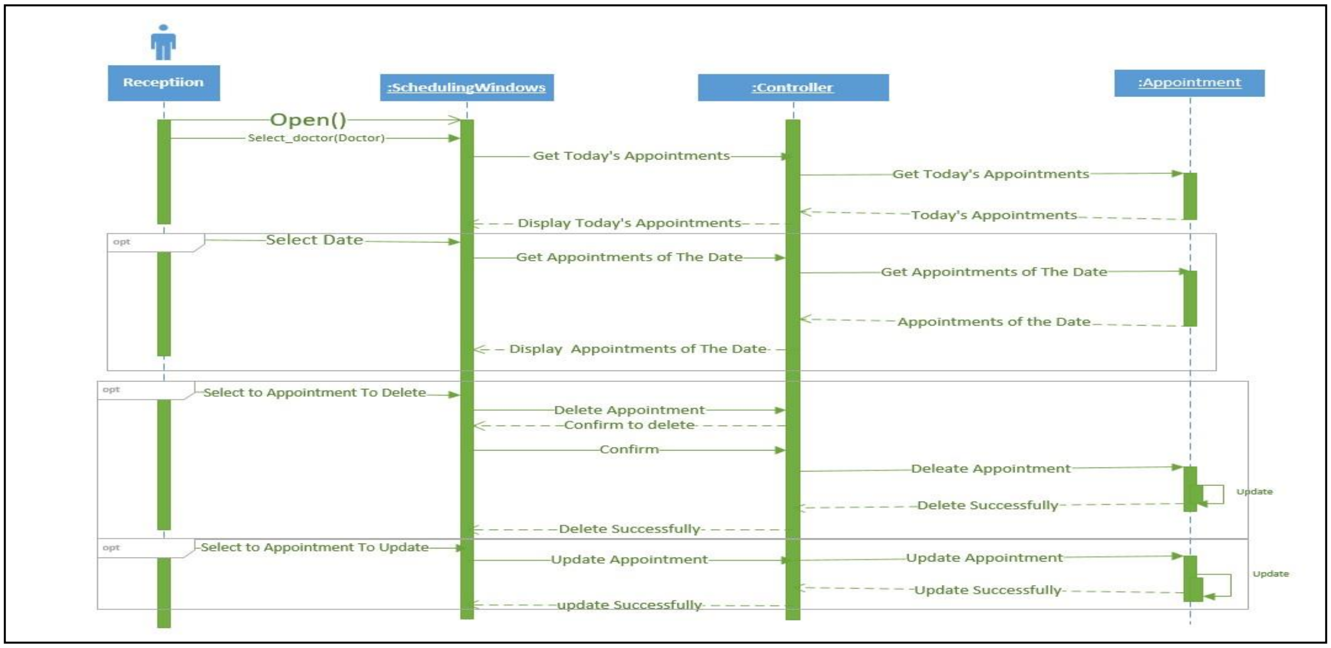
6. Other Requirements

This system depends 90% on databases, which means that it is necessary to properly design the database, as well as design the tables and their fields correctly, and also the link between the tables must be correct and effective, because any defect in the database may cause us big problems in the system

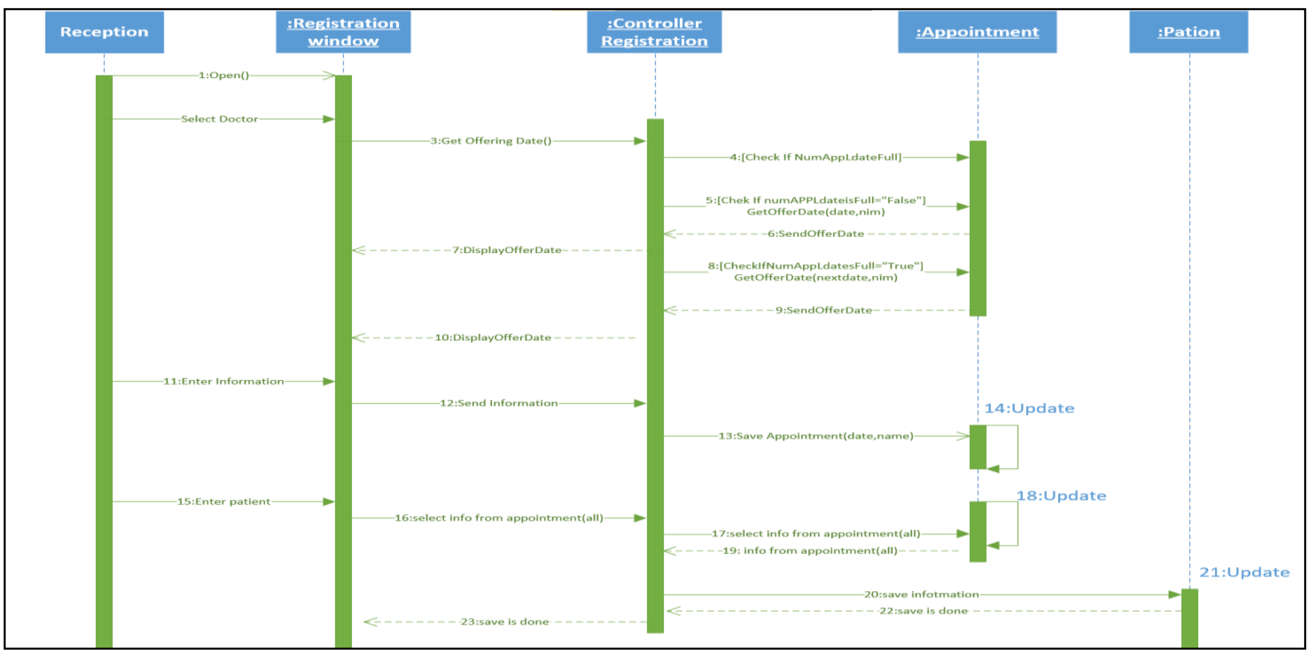
Appendix A: Analysis Models



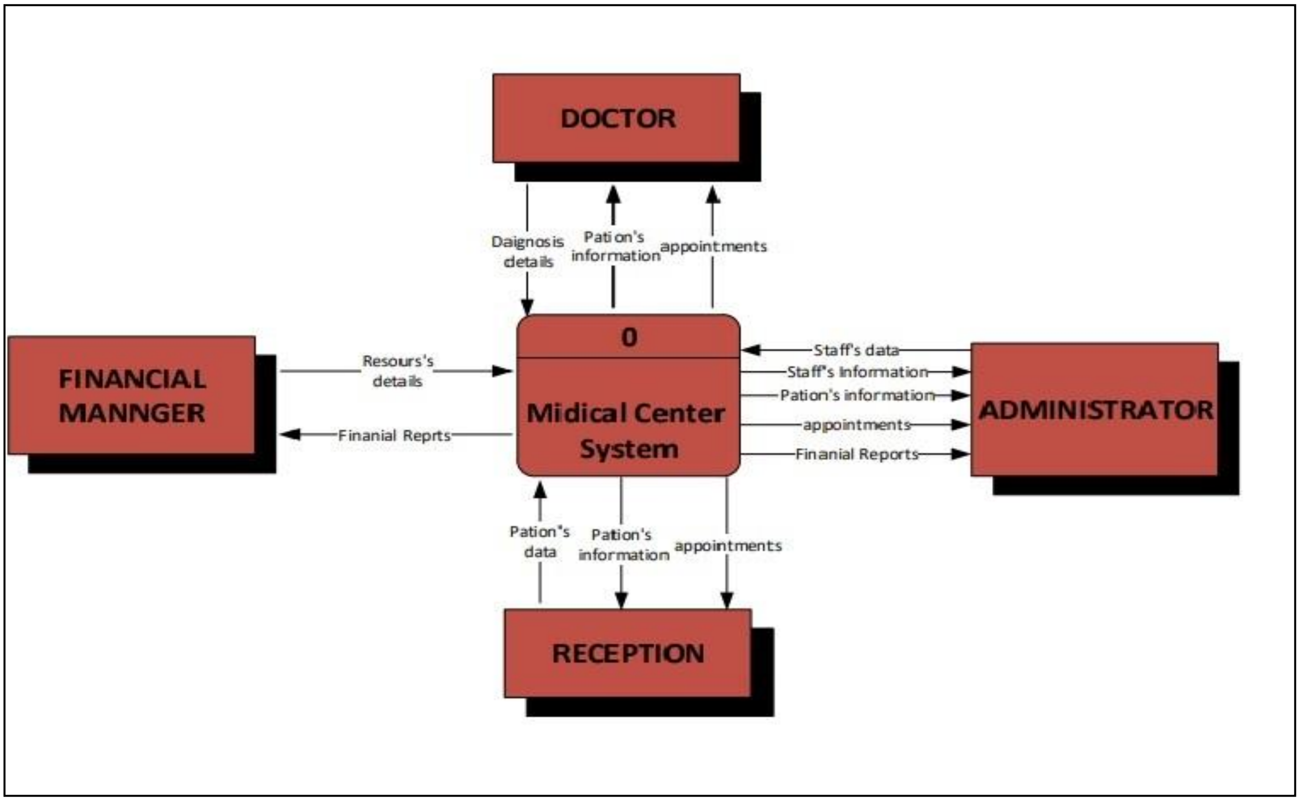
Figure(7) : the sequence diagram to mange the diagnoses from doctor



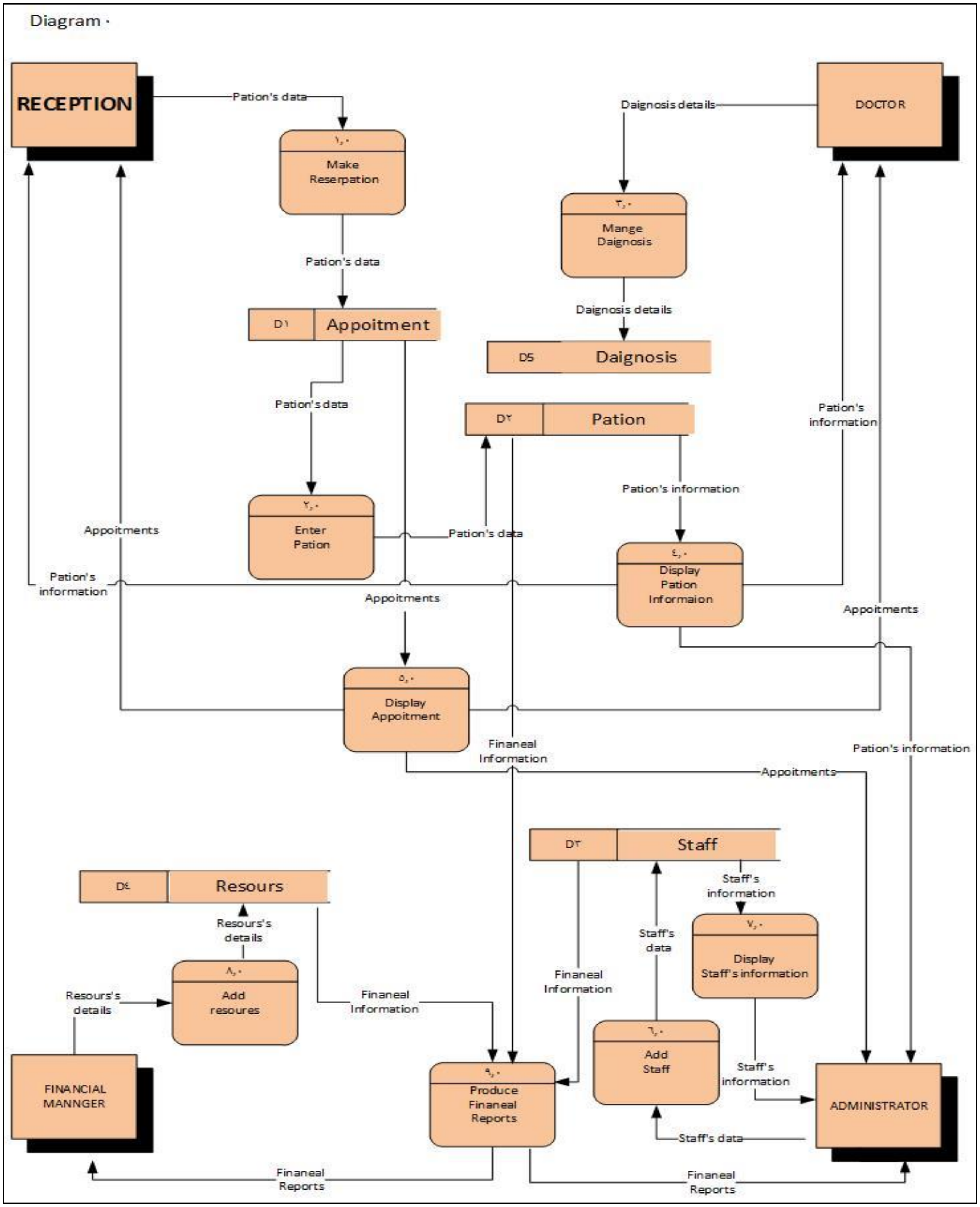
Figure(8) : sequence diagram to mange the appointments from reception



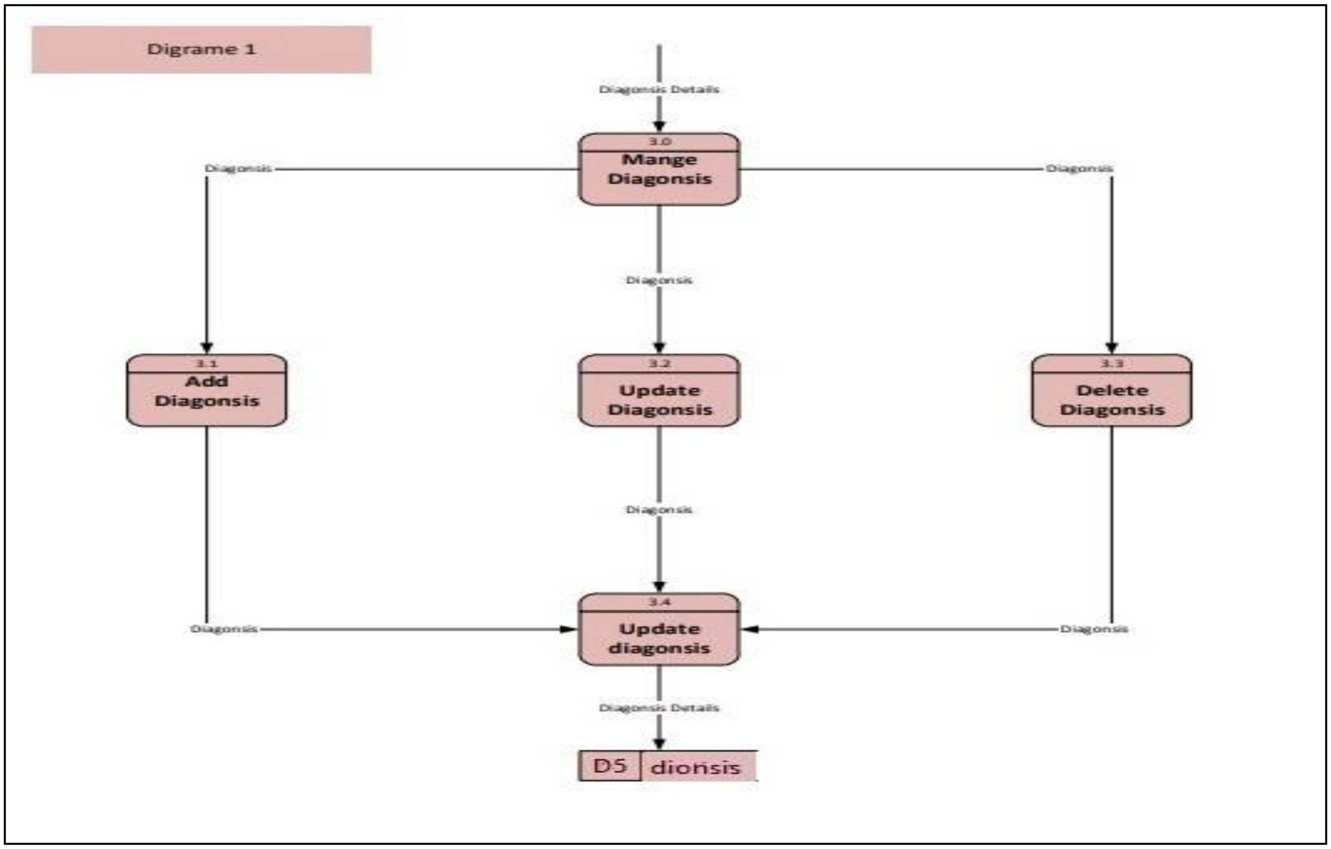
Figure(9) : sequence diagram to add patios data in system and enter patios to doctor



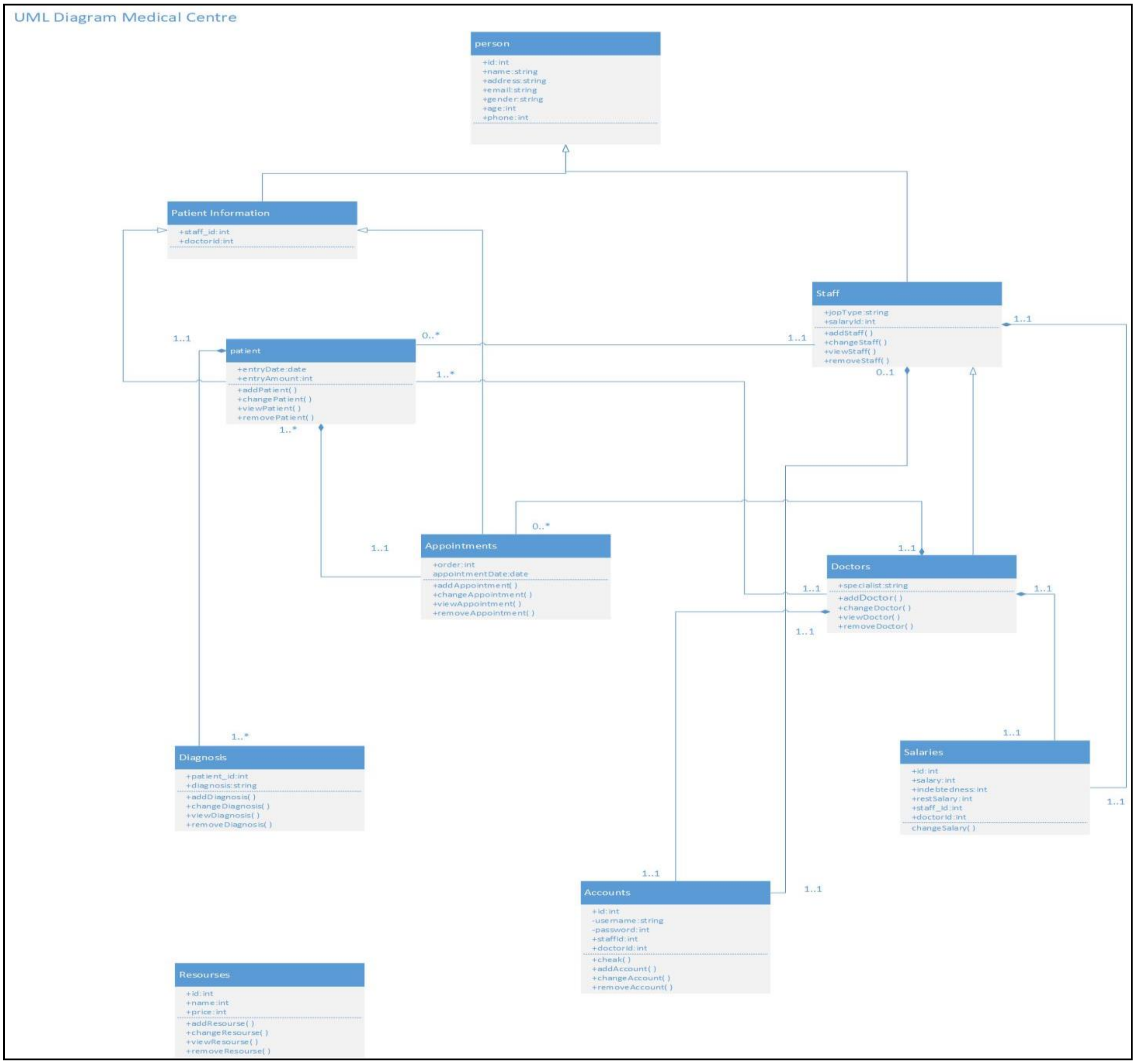
Figure(10) : context diagram for describes medical center process



Figure(11) : diagram 0 for some process in the system



Figure(12) : diagram 1 for process manage diagnosis



Figure(13) : UML diagram medical center